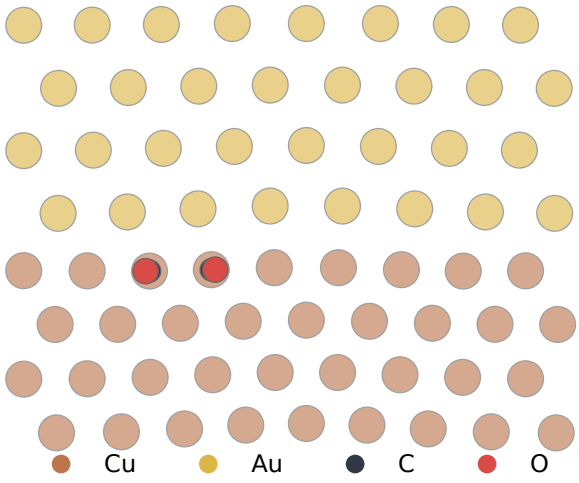
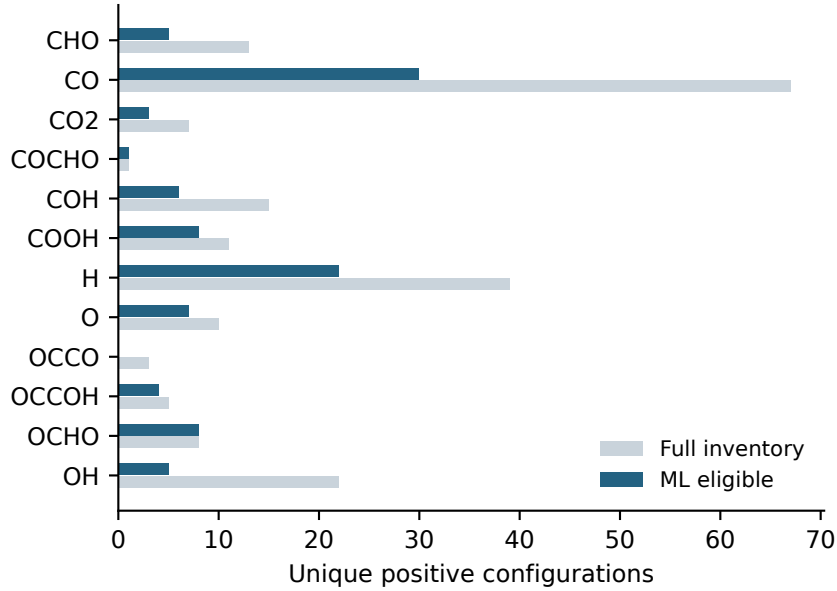
a Cu–Au interface input



b Species inventory



c Independent dataset controls

Parent
Pure metal or matching fcc(111) pair

Region
Interface · neighboring row · terrace

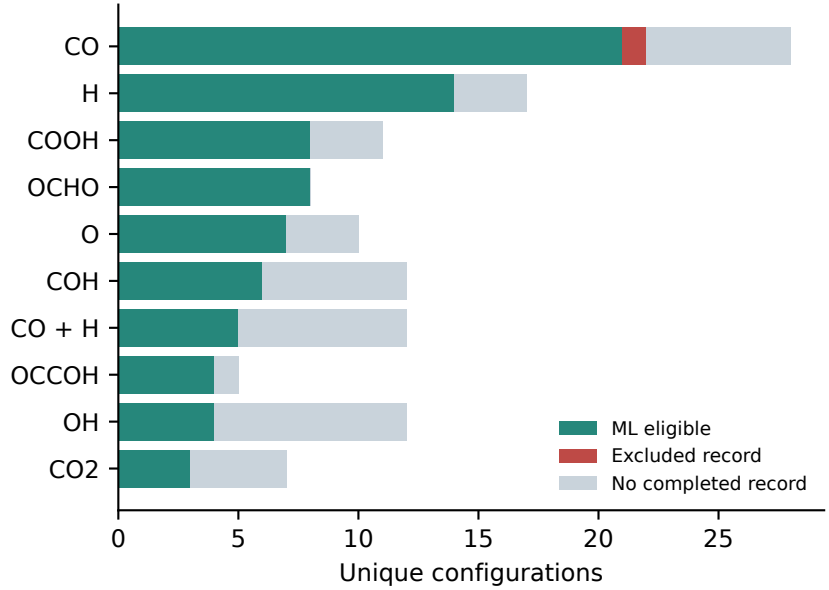
Binding
Atop · bridge · multicoordinated

Coverage
Molecules / declared surface reference

Distribution
Uniform · biased · coadsorbed

Chemistry
Verify components after relaxation

d Ten most populated identity sets



155 unique inventory structures; 87 ML eligible. Species counts overlap for coadsorption; identity sets are exclusive.